

Latch storage and state history

Two small explanations together: intentional latch storage and why current water-level inputs cannot replace state history.

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Reading days group the explanations, including later revisits. Tracker day labels and original dated submission evidence remain unchanged.

Use the linked contents or PDF bookmarks. Page numbers match the PDF viewer. Source questions and technical evidence remain beside their explanations.

11. D Latch: what 'latch inference' means

Question found in the successful Exams/m2014_q4a code

What does latch inference mean for these assignment lines, and why does the synthesis tool report a latch?

Answer

When `ena` is 0, the code requires `q` to retain its previous value. Retaining a value requires storage. Because the control is a level signal rather than a clock edge, synthesis creates a level-sensitive D latch. The warning is expected here because the HDLBits problem intentionally asks for a latch.

```
// Intentional latch: no assignment while ena is low.
always @(*) begin
    if (ena)
        q <= d;
end

// This says the same thing, but q <= q is redundant:
// else q <= q;
```

How the inferred hardware behaves

ena	Latch behavior	Meaning
1	q follows d	The latch is transparent while enabled.
0	q holds its old value	The latch is closed and stores the last d value.

Why this matters in ordinary combinational logic

In a block meant to be purely combinational, every output must be assigned on every path. If one path leaves an output unchanged, a latch can be inferred accidentally. Here it is intentional; elsewhere it is often a bug. In SystemVerilog, `always_latch` can state the intent more explicitly.

Latch versus flip-flop

A latch is **level-sensitive**: while enabled, output can respond to changes in `d`. A flip-flop is **edge-triggered**: output changes only at its specified clock edge.

Recall

No assignment (or `q <= q`) means 'remember the old value.' Remembering under a level enable is exactly the job of a latch.

https://hdlbits.01xz.net/wiki/Exams/m2014_q4a

Inputs are not a substitute for FSM state

The water sensors report the present level band. The state register stores any history that the outputs still need.

Original submitted-code question

```
//localparam s2 // i dont need the check for specific cases cause they are already in the
input
```

Answer

The input bits and the FSM state have different jobs. A sensor pattern tells the circuit the level **now**. A **localparam** does not test that pattern; it gives a readable binary encoding to an internal state. The next-state logic must still interpret the sensor pattern, and the state register must retain any past information needed by a Moore output.

Item	What it represents	Why it is needed
Sensor input	Current water-level band	Chooses the next transition at this clock
localparam state name	A constant binary state encoding	Makes the state machine readable and synthesizable
state register	Remembered operating condition/history	Lets Moore outputs depend only on registered state
case/if checks	Mapping from sensor patterns to next state	Defines behavior for every valid input and a safe default

Why the present input can be insufficient

In this problem, dfr remembers whether the level last moved down or up while it stays in a middle band. The same sensor pattern can therefore need different dfr values. Separate states preserve that history.

```
localparam [2:0] FROM_ABOVE = 3'b001,
                FROM_BELOW = 3'b010,
                HIGH_LEVEL = 3'b100;

always @(*) begin
    next_state = state;           // safe default
    if (highest_sensor)
        next_state = HIGH_LEVEL;
    else if (middle_band)
        next_state = came_from_high ? FROM_ABOVE : FROM_BELOW;
    else
        next_state = FROM_BELOW;
end
```

Important distinction

You may simplify checks for physically impossible sensor patterns if the prompt guarantees valid inputs. You may not remove a state when its remembered history changes an output.